

i_a (pu)
pre-fault
(load current)

Fault window

$$i_k(t) = \underbrace{[I_{ss} + (I_{sub} - I_{ss})e^{-t_{rel}/\tau_{ac}}]}_{\text{AC envelope}} \sin(\omega t + \phi_k)$$

$$+ \underbrace{I_{dc,k} e^{-t_{rel}/\tau_{dc}}}_{\text{DC offset}}, \quad t \geq t_0$$

$$I_{dc,k} = -I_{sub} \sin \phi_k \quad (\text{continuity cond.})$$

$$\phi_b = \phi_a - 2\pi/3, \quad \phi_c = \phi_a + 2\pi/3$$

$$I_{ss} + (I_{sub} - I_{ss})e^{-t_{rel}/\tau_{ac}}$$

$$I_{ss} \sin(\omega t + \phi_a)$$

t_0 : fault inception
 $i_a(t_0) = 0$ (continuity)

τ_{ac}
(AC decay)

$$I_{dc} = -I_{sub} \sin \phi_a$$

$$I_{dc} e^{-t_{rel}/\tau_{dc}}$$

- AC steady-state
- ... Envelope
- DC offset
- Total $i_a(t)$

T

t

t_0

$t_0 + T$

$t_0 + 2T$

$t_0 + 3T$

$t_0 + 4T$